



Kinetic Energy and Potential Energy

Kinetic Energy

Kinetic Energy is associated with the motion of an object. It depends on the mass and velocity of the object. A bullet travelling towards a target, a javelin moving through the air and a person running all have kinetic energy.
An object's Kinetic Energy (KE) can be calculated using the formula

Kinetic Energy(J) = ½ mass(kg) x velocity ^{2}(m/s)

KE = ½ mv^2

Example

Calculate the KE of a football which has a mass of 0.4 kg travelling at 5m/s

Formula

Working

Answer/ Units

Potential Energy

Energy can be stored in a variety of ways such as in food, as stored energy for our body, in petrol as stored energy for a car, in water turning the turbines to create electrical energy or a skier at the top of a mountain has stored energy which becomes kinetic energy (for movement)
An object's Potential Energy (PE) can be calculated using the formula

Potential Energy(J) = mass (kg) x gravity(9.8m/s/s) x height(m)

PE = mgh

Example

Calculate the PE of a tennis ball which has a mass of 0.05 kg hit to a height of 8m

Formula

Working

Answer/ Units

Kinetic energy and potential energy are related. Imagine a high board diver plunging straight down into the water. At the moment immediately before she leaves the board, her potential energy will be at a maximum. At this stage the diver is stationary and has zero kinetic energy but her position, high above the pool means her potential energy is high. When the diver leaves the board her KE increases until the instant prior to entry to the water when KE is at a maximum. At this point, PE is at a minimum since her height is now reduced to zero.



Explain how energy is converted in a snow skier.

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